SUPERELLIPTIC SELMER GROUPS

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1. The problems

Definition 1.0.1. Let k be a field. (♠ TODO: projective closure, plane curve, affine equation) (♠ TODO: smooth completion of an affine curve) A *superelliptic curve over k* is the smooth completion of the plane curve with affine equation of the form

$$y^d = f(x)$$

where $f(x) \in k[x]$ and $d \geq 2$ is some integer.

Definition 1.0.2. Let k be a field. Let C be a superelliptic curve (Definition 1.0.1) over k given as the smooth completion of the plane curve with affine equation

$$y^d = f(x)$$

where $f(x) \in k[x]$ and $d \geq 2$ is some integer. Given an element $u \in k^\times$, the *μ_d-twist of C* is the superelliptic curve given as the projective closure of the plane curve with affine equation

$$y^d = \frac{1}{u} f(x).$$

Here is the main problem:

Context 1.0.3. Let $K = \mathbb{F}_q(t)$ be a global function field of characteristic coprime to 6. Let $\ell \geq 5$ be a prime number such that $\gcd(\ell, q) = 1$. Write $F_{n,\ell}$ for the set of ℓth power free polynomials over $\mathbb{F}_q$ of degree n. Let v be a prime number coprime to $6q\ell$.

Context 1.0.4. Assume Context 1.0.3. Fix a superelliptic curve (Definition 1.0.1) C/K whose affine equation is given by

$$C : y^\ell = f(x)$$

for some non-trivial polynomial $f(x) \in K[x]$.

Given an ℓ-th power free polynomial $g \in \mathbb{F}_q[t]$ write C_g for the μ_ℓ-twist (Definition 1.0.2) of C by g. In other words C_g is the smooth completion of the plane curve with affine equation of the form

$$C_g : gy^\ell = f(x)$$

Problem 1.0.5. Assume Context 1.0.3 and Context 1.0.4

For each integer $j \geq 0$, what is the asymptotic value of the quantity

$$\frac{\#\{g \in F_{n,\ell} : \dim_{\mathbb{F}_v} \mathrm{Sel}_v(\mathrm{Jac}(C_g)/K) = j\}}{\#F_{n,\ell}}$$

(Definition D.0.2) (Definition A.1.3) as $n \to \infty$?

Here are subproblems

Problem 1.0.6.

Problem 1.0.7. Assume Context 1.0.3.

Write $\mathcal{J}_g$ for the Néron model of $\mathrm{Jac}(C_g)$ over

2. Hall's big monodromy approach

Theorem 2.0.1 (See e.g. [?, Chapter V Corollary 16.4])**.** Let $\mathbb{F}_q$ be a finite field of characteristic p with q elements, let G be a connected algebraic group over $\mathbb{F}_q$, and let F be the relative qth power Frobenius endomorphism on $G/\mathbb{F}_q$. The Lang map associated to F is a separable Galois étale (surjective) isogeny whose kernel is precisely $G(\mathbb{F}_q)$, the group of $\mathbb{F}_q$-points of G. (♠ TODO: TODO: Borel's book states the Lang isogeny as separable, and surjective, but I don't know what shows the Galoisness and étaleness)

Definition 2.0.2. Let $G/\mathbb{F}_q$ be a conected commutative algebraic group. Let $\ell \neq \mathrm{Char}\,\mathbb{F}_q$ be a prime number. Let $n \geq 1$. By Lang's theorem (Theorem 2.0.1), given a connected commutative algebraic group $G/\mathbb{F}_q$, we have we have a short exact sequence

$$1 \to G(\mathbb{F}_{q^n}) \to G_{\mathbb{F}_{q^n}} = G \times_{\mathbb{F}_q} \mathbb{F}_{q^n} \xrightarrow{L} G_{\mathbb{F}_{q^n}} \to 1$$

where L is the Lang isogeny. Since this identifies $G_{\mathbb{F}_{q^n}}$ with a finite étale $G(\mathbb{F}_{q^n})$-cover of $G_{\mathbb{F}_{q^n}}$, we have a corresponding surjective group homomorphism $\pi_1^{\text{ét}}(G_{\mathbb{F}_{q^n}}, e) \to G(\mathbb{F}_{q^n})$.

1. Let $\chi : G(\mathbb{F}_{q^n}) \to \mathbb{F}_\ell^\times$ be a character. There is a lisse sheaf $\mathcal{L}_\chi$ of rank 1 with coefficients in $\mathbb{F}_\ell$ on $G_{\mathbb{F}_{q^n}} = G \times_{\mathbb{F}_q} \mathbb{F}_{q^n}$ corresponding (Theorem E.0.2) to the composed group homomorphism
$$\pi_1^{\text{ét}}(G_{\mathbb{F}_{q^n}}, e) \to G(\mathbb{F}_{q^n}) \xrightarrow{\chi} \mathbb{F}_\ell^\times.$$
We may call this lisse sheaf the *Kummer sheaf associated to* χ.

   Similarly, Kummer sheaves are associated to characters $\chi : G(\mathbb{F}_{q^n}) \to \mathbb{Z}_\ell^\times$, $\chi : G(\mathbb{F}_{q^n}) \to \mathbb{Q}_\ell^\times$, $\chi : G(\mathbb{F}_{q^n}) \to \overline{\mathbb{Q}}_\ell^\times$, etc.; each type of Kummer sheaf is a lisse sheaf on $G_{\mathbb{F}_{q^n}}$ with appropriate coefficients.

Definition 2.0.3. Let X be an integral normal scheme with generic point η and function field K. Let $\overline{\eta}$ be a geometric point of X whose image is η. Let $U \subseteq X$ be a non-empty open subscheme. Let $x \in X \setminus U$ be a point. Let v be a valuation of K centered at x.

Recall that there is a natural surjection
$$\operatorname{Gal}(K^{\text{sep}}/K) \twoheadrightarrow \pi_1^{\text{ét}}(U, \overline{\eta})$$
(Proposition E.0.5)

1. The *decomposition subgroup* D_v of $\pi_1^{\text{ét}}(U, \overline{\eta})$ is the image of the decomposition subgroup (Definition E.0.3) $D_{K,v}$ of $\operatorname{Gal}(K^{\text{sep}}/K)$ in $\pi_1^{\text{ét}}(U, \overline{\eta})$.
2. The *inertia subgroup* I_v of $\pi_1^{\text{ét}}(U, \overline{\eta})$ is the image of the inertia subgroup (Definition E.0.4) $I_{K,v}$ of $\operatorname{Gal}(K^{\text{sep}}/K)$ in $\pi_1^{\text{ét}}(U, \overline{\eta})$.

Given two choices v_1 and v_2 of valuations centered at x, the groups D_{v_1} and D_{v_2} and the groups I_{v_1} and I_{v_2} are conjugate to each other in $\pi_1^{\text{ét}}(U, \overline{\eta})$. As such, one might write D_x and I_x for the conjugacy classes.

Definition 2.0.4. Let X be an connected normal scheme with generic point η and function field K. Let $\overline{\eta}$ be a geometric point of X whose image is η. Let $U \subseteq X$ be a non-empty open subscheme. Let $x \in X \setminus U$ be a point.

Let $\mathcal{F}$ be one of the following:

1. A finite locally constant sheaf of sets on $U_{\text{ét}}$
2. A finite locally constant sheaf of finite cardinality abelian groups on $U_{\text{ét}}$
3. A locally constant sheaf of finitely generated modules over a finite ring on $U_{\text{ét}}$
4. In the case that U is irreducible and geometrically unibranch, a finite type, locally constant sheaf of Λ-modules on $U_{\text{ét}}$
5. Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R.
   (a) A lisse object of $\operatorname{Sh}(U, R)$ (**??**)
   (b) A lisse object of $\operatorname{Sh}(U, E)$ (**??**)
   (c) For an algebraic extension K of E, a lisse object of $\operatorname{Sh}(U, K)$ (**??**).

In each of the above cases for $\mathcal{F}$, there is a corresponding action of $\pi_1^{\text{ét}}(U, \overline{\eta})$ on the stalk $\mathcal{F}_\eta$ (Theorem E.0.1) (Theorem E.0.2):
$$\pi_1^{\text{ét}}(U, \overline{\eta}) \to \operatorname{Aut}(\mathcal{F}_\eta)$$

where $\mathrm{Aut}(\mathcal{F}_\eta)$ is the automorphism group for the appropriate category that $\mathcal{F}_\eta$ belongs to. Further recall that there is an inertia subgroup (Definition 2.0.3) I_x of $\pi_1^{\text{ét}}(U, \overline{\eta})$, well defined up to conjugation.

We say that $\mathcal{F}$ is *unramified at x* if the action of I_x on $\mathcal{F}_{\overline{\eta}}$ is trivial, i.e. the composed homomorphism

$$I_x \hookrightarrow \pi_1^{\text{ét}}(U, \overline{\eta}) \to \mathrm{Aut}(\mathcal{F}_\eta)$$

is trivial. Otherwise, we say that $\mathcal{F}$ is *ramified at x*.

Let $C/\mathbb{F}_q$ be a smooth, geometrically connected curve. Let $X/K(C)$ be a superelliptic curve with affine model $y^r = f(x)$ for a polynomial $f(x) \in K(C)[x]$. Given $g \in K(C)^\times$, let X_g be the twist of X with affine model $g \cdot y^r = f(x)$. Write $\mathcal{X}_g \to C$ for the Néron model of X_g. For a prime number v not dividing $\mathrm{Char}\,\mathbb{F}_q$, $\mathrm{Pic}^0_{\mathcal{X}_g/C}[v]$ is a quasi-finite étale group scheme over C (♠ TODO: cite).

Theorem 2.0.5 (Explicit construction of $\mathcal{T}_{d,\ell}$ as a derived pushforward [**?**, Section 5])**.** Let q be a prime power not divisible by $2, 3$. Let $C/\mathbb{F}_q$ be a proper smooth geometrically connected curve and write $K = \mathbb{F}_q(C)$ for its function field. Fix an elliptic curve E_1/K with non-constant j-invariant and let $\mathcal{E}_1 \to C$ be its Néron model (Definition B.0.4). Let ℓ be a prime invertible in K. Write $\mathcal{E}_{1,\ell} \subset \mathcal{E}_1$ for the kernel of multiplication by ℓ; this is a quasi-finite étale group scheme over C.

For every positive integer d, let F_d be the parameter space of square-free polynomials of degree d relatively prime to a fixed polynomial $m \in \mathbb{F}_q[t]$. Concretely, if $C = \P^1_{\mathbb{F}_q}$ and $K = \mathbb{F}_q(t)$, then

$$F_d = \mathrm{Spec}(\mathbb{F}_q[c_0, c_1, \ldots, c_d]) \setminus (\Delta \cup V(m)),$$

where Δ is the discriminant locus and $V(m)$ enforces coprimality with m. For each geometric point $\bar{f} \in F_d(\overline{\mathbb{F}}_q)$ corresponding to a polynomial f, let E_f/K denote the quadratic twist of E_1/K by f. Let $\mathcal{E}_f \to C$ be its Néron model and $\mathcal{E}_{f,\ell} \subset \mathcal{E}_f$ the kernel of multiplication by ℓ.

Let $P(t, c) = t^d + c_{d-1}t^{d-1} + \cdots + c_1 t + c_0 \in \mathbb{F}_q[t][c_0, \ldots, c_d]$ be the universal monic polynomial of degree d. Consider the base scheme

$$S := C \times F_d.$$

Let $\mathcal{E}_{\mathrm{univ}} = \mathcal{E}_1 \times_C S \to S$ be the pullback of the Néron model along the projection $S \to C$.

Let $W' = \mathbb{A}^1_v \times S$ with coordinates (v, t, c), and let $\widetilde{S} \subset W'$ be the closed subscheme defined by

$$v^2 = P(t, c).$$

The map $\sigma : \widetilde{S} \to S$, $(v, t, c) \mapsto (t, c)$, is a finite flat double cover. Let $U_{\mathrm{ram}} \subset S$ be the ramification locus, defined by $P(t, c) = 0$, and set $U = S \setminus U_{\mathrm{ram}}$. The restriction $\sigma|_{\widetilde{S}|_U} : \widetilde{S}|_U \to U$ is a finite étale Galois cover with Galois group $\mathbb{Z}/2\mathbb{Z}$.

Let $\iota : \widetilde{S} \to \widetilde{S}$ be the deck involution, $(v, t, c) \mapsto (-v, t, c)$. Let $[-1]_{\mathcal{E}} : \mathcal{E}_{\mathrm{univ}} \to \mathcal{E}_{\mathrm{univ}}$ denote fiberwise negation on the elliptic curve. Pull back $\mathcal{E}_{\mathrm{univ}}$ to $\widetilde{S}$ via σ to obtain $\sigma^*\mathcal{E}_{\mathrm{univ}} \to \widetilde{S}$.

The group $\mathbb{Z}/2\mathbb{Z}$ acts on $\sigma^*\mathcal{E}_{\text{univ}}$ via the combined action

$$(v, x) \mapsto (\iota(v), [-1]_{\mathcal{E}}(x)) = (-v, -x),$$

where x denotes a point of the elliptic curve fiber. This action is étale and free on the generic fiber.

The descent quotient

$$\mathcal{G} := \sigma^*\mathcal{E}_{\text{univ}}/\langle(\iota, -1)\rangle$$

is an elliptic curve over S.

To see that its fiber over $(t_0, c_0) \in S$ is the quadratic twist of $\mathcal{E}_1$ by $P(t_0, c_0)$, compute with Weierstrass equations. Since 2 is invertible in K, the Néron model $\mathcal{E}_1 \to C$ admits a relative Weierstrass equation

$$y^2 = x^3 + Ax + B$$

over C, where A, B are sections of appropriate powers of the relative dualizing sheaf. Let $p : S = C \times F_d \to C$ be the projection and $\sigma : \widetilde{S} \to S$ the double cover. Pulling back to $\widetilde{S}$ via $\sigma \circ p$, the total space $\sigma^*\mathcal{E}_{\text{univ}} \to \widetilde{S}$ has equation

$$y^2 = x^3 + (\sigma \circ p)^*A \cdot x + (\sigma \circ p)^*B.$$

The involution $(\iota, -1)$ acts on the coordinates of $\sigma^*\mathcal{E}_{\text{univ}}$ as $(v, x, y) \mapsto (-v, x, -y)$: it fixes x and negates both v and y. The quotient scheme is obtained by taking invariants. Setting

$$X = x, \quad Y = vy,$$

which are invariant under the action since $(-v)(-y) = vy$, the equation of the descended curve $\mathcal{G} \to S$ becomes

$$Y^2 = v^2y^2 = P(t, c)\left(X^3 + (\sigma \circ p)^*A \cdot X + (\sigma \circ p)^*B\right).$$

Since $P(t, c)$, $(\sigma \circ p)^*A$, and $(\sigma \circ p)^*B$ are all pullbacks from S (they do not depend on v), this equation descends to S. On S it reads

$$Y^2 = P(t, c)\left(X^3 + p^*A \cdot X + p^*B\right).$$

This is the quadratic twist of $\mathcal{E}_{\text{univ}}$ by $P(t, c)$, given in the model produced by Galois descent. Restricting to any fiber $C \times \{\bar{f}\}$ where $c \mapsto f$, this gives the quadratic twist of $\mathcal{E}_1$ by $f \in K^\times/(K^\times)^2$.

Thus the fiber of $\mathcal{G}$ over $(t, c) \in S$ is the quadratic twist of the fiber of $\mathcal{E}_1$ at t by the value $P(t, c) \in K^\times/(K^\times)^2$. In particular, for each fixed $\bar{f} \in F_d(\overline{\mathbb{F}}_q)$ with corresponding polynomial f, the restriction of $\mathcal{G}$ to $C \times \{\bar{f}\}$ is canonically isomorphic to $\mathcal{E}_f \to C$.

Let $\mathcal{G}_\ell \subset \mathcal{G}$ be the kernel of multiplication by ℓ. This is a finite étale group scheme over S. Let $j : S_{\text{sm}} \hookrightarrow S$ be the open immersion of the locus where $\mathcal{G} \to S$ has good or multiplicative reduction (the complement of the additive reduction locus). The middle extension

$$j_*j^*\mathcal{G}_\ell$$

is a constructible $\mathbb{F}_\ell$-sheaf on S. By assumption on ℓ, $\mathcal{G}_\ell$ is already a middle extension over each fiber $C \times \{\bar{f}\}$.

Let $\pi : S = C \times F_d \to F_d$ be the projection. The first higher direct image

$$\mathcal{T}_{d,\ell} := R^1\pi_*(j_*j^*\mathcal{G}_\ell)$$

is a lisse $\mathbb{F}_\ell$-sheaf on F_d.

For each geometric point $\bar{f} \in F_d(\overline{\mathbb{F}}_q)$, the geometric fiber is

$$\mathcal{T}_{d,\ell}|_{\bar{f}} \cong H^1(C \times \overline{\mathbb{F}}_q, \mathcal{E}_{f,\ell}),$$

where $\mathcal{E}_{f,\ell}$ is the Néron model of the ℓ-torsion $E_f[\ell]$ for the quadratic twist E_f/K.

Remark 2.0.6 (Orthogonal pairing)**.** The Weil pairing on $E_f[\ell] \times E_f[\ell] \to \mu_\ell$ extends to a non-degenerate alternating pairing $\mathcal{E}_{f,\ell} \times \mathcal{E}_{f,\ell} \to \mathbb{F}_\ell(1)$, making $\mathcal{E}_{f,\ell}$ self-dual. Together with Poincaré duality this induces a non-degenerate symmetric (orthogonal) pairing on the cohomology

$$H^1(C \times \overline{\mathbb{F}}_q, \mathcal{E}_{f,\ell}) \times H^1(C \times \overline{\mathbb{F}}_q, \mathcal{E}_{f,\ell}) \longrightarrow \mathbb{F}_\ell.$$

The sheaf $\mathcal{T}_{d,\ell}$ carries this orthogonal pairing fiberwise, so the monodromy group of $\mathcal{T}_{d,\ell}$ lies in the orthogonal group $O(V_{f,\ell})$.

Definition 2.0.7.

Appendix A. Abelian varieties

Definition A.0.1. Let k be a field. A *(algebraic) variety over k* is an integral, separated scheme of finite type over k.

Definition A.0.2. Let S be a scheme. An algebraic group scheme over S (or an S-group scheme) is a group object G in the category of schemes over S; that is, G is an S-scheme equipped with S-morphisms: $m : G \times_S G \to G$ (*multiplication*), $i : G \to G$ (*inverse*), and $e : S \to G$ (*identity*/*identity section*), satisfying the group axioms expressed by the commutativity of the following diagrams:

1. **Associativity**
$$\begin{array}{ccc} G \times_S G \times_S G & \overset{m\times\mathrm{id}}{\longrightarrow} & G \times_S G \\ {\scriptstyle \mathrm{id}\times m}\downarrow & & \downarrow{\scriptstyle m} \\ G \times_S G & \xrightarrow{\ m\ } & G \end{array}$$

2. **Identity**
$$\begin{array}{ccc} G \times_S S & \overset{\mathrm{id}\times e}{\longrightarrow} & G \times_S G \\ & {\scriptstyle \simeq}\searrow & \downarrow{\scriptstyle m} \\ & & G \end{array} \qquad \begin{array}{ccc} S \times_S G & \overset{e\times\mathrm{id}}{\longrightarrow} & G \times_S G \\ & {\scriptstyle \simeq}\searrow & \downarrow{\scriptstyle m} \\ & & G \end{array}$$

3. **Inverse**
$$\begin{array}{ccc} G & \overset{(\mathrm{id},i)}{\longrightarrow} & G \times_S G \\ {\scriptstyle \mathrm{id}}\downarrow & & \downarrow{\scriptstyle m} \\ G & \xrightarrow{\ e\circ\pi\ } & G \end{array}$$
where $\pi : G \to S$ is the structure morphism and $e \circ \pi$ sends g to the identity section.

If the multiplication map also satisfies the commutative axiom, then the group scheme G is called *commutative* or *abelian*.

If G is affine over S, we call it an *affine group scheme over S*.

If the base scheme S is the spectrum of a field k, then we call G a *k-algebraic group* or an *algebraic group (scheme) over k*. If G is additionally a k-variety, then we call G a *k-group variety*.

Definition A.0.3. Let S be a scheme, and let G and H be S-algebraic groups (Definition A.0.2). A morphism of S-schemes $f : G \to H$ is a *homomorphism of algebraic groups* if f is a group homomorphism, i.e.,

- $f(m_G(x, y)) = m_H(f(x), f(y)) \quad$ for all $x, y \in G$,
- $f(i_G(x)) = i_H(f(x))$ for all $x \in G$, and
- $f(e_G) = e_H$.

It is called an *isomorphism of algebraic groups (over S)* if it is additionally an isomorphism of S-schemes. If there exists an isomorphism $f : G \to H$ of algebraic groups over S, then G and H are said to be *isomorphic S-algebraic groups*.

Definition A.0.4 (Kernel of a homomorphism of group schemes over a scheme)**.** Let S be a scheme and let $\phi\colon G_1 \to G_2$ be a homomorphism of algebraic group schemes over S (Definition A.0.3).

The *kernel of ϕ*, denoted by $\ker \phi$ or $A[\phi]$, is the fiber of ϕ at the identity section (Definition A.0.2) $e_2\colon S \to G_2$ of G_2. Equivalently, $\ker \phi$ is the subgroup scheme of G_1 defined as the pullback:

$$\ker \phi = G_1 \times_{G_2} S,$$

(♠ TODO: finite group scheme) where the map $G_1 \to G_2$ is ϕ and the map $S \to G_2$ is the identity section. If ϕ is a finite morphism, then $\ker \phi$ is a finite group scheme over S.

(♠ TODO: group variety)

Definition A.0.5 (Abelian variety over a field)**.** Let k be a field. An *abelian variety over k* is a complete, connected algebraic group variety defined over k, i.e.

- A is a smooth, proper, geometrically connected algebraic variety over k,
- A is endowed with a group structure defined by morphisms of varieties over k (multiplication $m : A \times_k A \to A$ and inverse $i : A \to A$),
- the group law satisfies the group axioms scheme-theoretically.

In particular, an abelian variety is a projective algebraic group variety (Definition A.0.2) over k.

Definition A.0.6 (Isogeny of abelian varieties over a field)**.** (♠ TODO: finite group scheme) Let k be a field and let A and B be abelian varieties (Definition A.0.5) over k. An *isogeny of abelian varieties over k* is a homomorphism (Definition A.0.3) $\varphi\colon A \to B$ of group schemes over k that is surjective and has finite kernel (Definition A.0.4). The kernel $\ker \varphi$ is a finite group scheme over k (Definition A.0.4). Equivalently, φ is a finite, faithfully flat homomorphism of abelian varieties over k.

A.1. **Picard groups and Jacobians.**

Definition A.1.1. (♠ TODO: picard group of an algebraic space) Let X be a scheme. The *Picard group of X*, denoted by $\mathrm{Pic}(X)$, is the group of isomorphism classes of invertible sheaves (line bundles) on X (with respect to the Zariski topology), with the group operation induced by the tensor product of sheaves.

More precisely:

1. Two invertible sheaves $\mathcal{L}$ and $\mathcal{M}$ are isomorphic if there exists an $\mathcal{O}_X$-module isomorphism $\phi : \mathcal{L} \xrightarrow{\sim} \mathcal{M}$.
2. The tensor product of invertible sheaves induces a well-defined group operation on the set of isomorphism classes. (♠ TODO: sheafy dual)
3. The inverse of an invertible sheaf $\mathcal{L}$ is given by its dual $\mathcal{L}^\vee = \mathscr{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$.
4. The identity element is the class of the structure sheaf $\mathcal{O}_X$.

(♠ TODO: class group of ring) When $X = \mathrm{Spec}(A)$ is an affine scheme, there is a canonical isomorphism

$$\mathrm{Pic}(\mathrm{Spec}(A)) \cong \mathrm{Cl}(A),$$

where $\mathrm{Cl}(A)$ is the class group of the ring A (the group of isomorphism classes of rank-1 projective modules over A, also known as the Picard group of the ring).

(♠ TODO: extract this out) If X is a Noetherian, integral, and locally factorial scheme, then $\mathrm{Pic}(X)$ is isomorphic to the Weil divisor class group $\mathrm{Cl}(X)$.

Definition A.1.2. Let X/k be a smooth, geometrically integral projective variety over a field k.

The relative Picard functor $\mathrm{Pic}_{X/k}$ is representable by a smooth scheme over k. (♠ TODO: show that the identity component is an abelian variety) The identity component $\mathrm{Pic}^0_{X/k}$ is in fact an abelian variety and is called the *Picard variety of X/k*.

Definition A.1.3 (Jacobian variety of a curve)**.** (♠ TODO: separate out jacobian of just a proper curve and the jacobian of a nice curve) Let X be a proper curve over a field k.

1. The relative Picard functor $\mathrm{Pic}_{X/k}$ is representable by a smooth group scheme (Definition A.0.2) over k. The *(generalized) Jacobian of X* refers to the identity component $\mathrm{Pic}^0_{X/k}$ of this group scheme.
2. Now assume that X is smooth, is projective, and has geometrically connected connected components. The Jacobian is an abelian variety (Definition A.0.5) over k and may be called the *Jacobian variety of X*. It may be denoted by notations such as $\mathrm{Jac}(X)$, Jac_X, or J_X.

   Under these assumptions, note that $\mathrm{Jac}(X)$ coincides with the Picard variety (Definition A.1.2) of X/k. Equivalently, J_X parametrizes isomorphism classes of degree-zero line bundles on X, where a line bundle $\mathcal{L}$ has *degree zero* if its degree (computed as the sum of orders of any rational function defining it) vanishes.

The Jacobian variety $\mathrm{Jac}(X)$ has dimension g where g is the geometric genus of X. When X is equipped with a base point, the Jacobian variety carries a canonical principal polarization Θ, making (J_X, Θ) a principally polarized abelian variety.

Appendix B. Néron models of abelian varieties

Definition B.0.1 (Dedekind domain)**.** An integral domain R is called a *Dedekind domain* or a *Dedekind ring* if it satisfies the following equivalent conditions:

- R is Noetherian, integrally closed in its field of fractions, and every nonzero prime ideal of R is maximal.
- Equivalently: for every nonzero prime ideal $\mathfrak{p}$ of R, the localization $R_{\mathfrak{p}}$ is a discrete valuation ring.

Definition B.0.2 (Dedekind scheme)**.** A *Dedekind scheme* is a scheme S that is integral, Noetherian, normal, and of Krull dimension 1. Equivalently, every point $s \in S$ has an open neighborhood $U = \operatorname{Spec} R$ such that R is a Dedekind domain (Definition B.0.1). In particular, for every point $s \in S$, the local ring $\mathcal{O}_{S,s}$ is either a field (if s is the generic point) or a discrete valuation ring (if s is a closed point).

Definition B.0.3 (Special fiber and generic fiber)**.** Let S be a integral scheme, let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S. Let $s \in S$ be a closed point.

1. For any morphism of schemes $f : X \to S$, the *generic fiber of X over S* (or simply the *generic fiber*) is the base change:

$$X_\eta = X \times_S \operatorname{Spec} K$$

of f along $\operatorname{Spec} K \to S$.
2. For any morphism of schemes $f : X \to S$, the *special fiber of X at s* is the base change:

$$X_s = X \times_S \operatorname{Spec} k(s)$$

of f along $\operatorname{Spec} k(s) \to S$, where $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$ is the residue field at s.

In the case that S has only closed point (e.g. S is the ring of integers of a local field), the *special fiber of X* refers to the special fiber of X at this unique closed point.

Definition B.0.4 (Néron model of an abelian variety)**.** Let S be an integral scheme, let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S. Let A_K be an abelian variety over K (Definition A.0.5).

A *Néron model of A_K over S* is a smooth separated group scheme $\mathcal{A}$ of finite type over S whose generic fiber (Definition B.0.3) $\mathcal{A}_\eta$ is isomorphic to A_K, satisfying the following universal property (the *Néron mapping property*): for every smooth S-scheme U and every K-morphism $f_K : U_\eta \to A_K$, there exists a unique S-morphism $f : U \to \mathcal{A}$ extending f_K.

If such a model exists, it is unique up to unique isomorphism.

Definition C.0.1. Let G be a group and R a not necessarily commutative ring. Let A be a left/right $R[G]$-module.

1. The *cohomology group of G with coefficients in A* or the *group cohomology of A as an $R[G]$-module*, denoted $H^n(G,A)$ or $H^n(G;A)$for $n \geq 0$, is the n-th right derived functor object $(R^n(-^G))(A)$ of the invariants functor $A \mapsto A^G$, which is left exact by **??**.
2. The *homology group of G with coefficients in A* or the *group homology of A as an $R[G]$-module*, denoted $H_n(G,A)$ or $H_n(G;A)$, for $n \geq 0$, is the n-th left derived functor object $(L_n(-_G))(A)$ of the coinvariants functor $A \mapsto A_G$, which is right exact by **??**.

Definition C.0.2. Let R be a topological ring and G a topological group. Let R-$\mathbf{TopMod}_G$ be the category of (left) topological G-modules over R.

Let M be an object in R-$\mathbf{TopMod}_G$. The *continuous cochain complex* $C^\bullet_{\text{cont}}(G,M)$ (also denoted by $C^\bullet(G,M)$, but which should not be confused with the cochain complex $C^\bullet(G,M)$ that does not incorporate any topological structure) is defined as follows:

1. $C^n_{\text{cont}}(G,M) = \text{Map}_{\text{cont}}(G^n, M)$, the R-module of continuous maps from G^n to M.
2. The *coboundary map* $d^n : C^n_{\text{cont}}(G,M) \to C^{n+1}_{\text{cont}}(G,M)$ is given by:

$$(d^n f)(g_1,\ldots,g_{n+1}) = g_1 f(g_2,\ldots,g_{n+1}) + \sum_{i=1}^{n} (-1)^i f(g_1,\ldots,g_i g_{i+1},\ldots,g_{n+1}) + (-1)^{n+1} f(g_1,\ldots,g_n)$$

Definition C.0.3. Let R be a topological ring and G a topological group. Let R-$\mathbf{TopMod}_G$ be the category of (left) topological G-modules over R.

The *continuous cohomology groups* $H^n_{\text{cont}}(G,M)$ are defined as the cohomology groups of the continuous cochain complex (Definition C.0.2) $C^\bullet_{\text{cont}}(G,M)$.

The continuous cohomology group is also denoted by $H^n(G,M)$ if the continuity is clear from context, but this should not be confused for the (non-continuous) cohomology group (Definition C.0.1) $H^n(G,M)$ of M as a G-module (without consideration of the topological structures).

Theorem C.0.4. (♠ TODO: verify these facts)

Let R be a topological ring and G a topological group. Let R-$\mathbf{TopMod}_G$ be the category of (left) topological G-modules over R.

Consider the G-invariant functor $(-)^G : R\text{-}\mathbf{TopMod}_G \to R\text{-}\mathbf{TopMod}$ (**??**).

Assume either of the following:

1. G is profinite, R is discrete, and M is discrete; in this case, by the "right derived functors of $(-)^G$", we mean the right derived functor on the category of discrete G-modules, which has enough injectives (♠ TODO: cite; apparently the category of

discrete G-modules over a topological ring R where G is profinite is an abelian category with enough injectives).
2. G is locally compact, and M is complete; in this case, by the "right derived functors of $(-)^G$", we mean the right derived functor in the sense of relative homological algebra, using resolutions that split as sequences of topological spaces (♠ TODO: figureo ut what this means).

The continuous cohomology groups (Definition C.0.3) $H^n_{\mathrm{cont}}(G, M)$, i.e. the cohomology of the continuous cochain complex (Definition C.0.2) $C^\bullet_{\mathrm{cont}}(G, M)$, is naturally isomorphic to the the right derived functors of $(-)^G$. In other words,

$$H^n(C^\bullet_{\mathrm{cont}}(G, M)) \cong (R^n(-^G))(M)$$

Definition C.0.5. Let R be a topological ring (e.g., a ℓ-adic ring $\mathbb{Z}_\ell$).

1. Let L/K be a Galois field extension. Let M be a Galois module of $\mathrm{Gal}(L/K)$ over R
   The *Galois cohomology group of L/K with coefficients in M* is:
   $$H^n(L/K, M) = H^n_{\mathrm{cont}}(\mathrm{Gal}(L/K), M)$$
   (Definition C.0.3).
2. Let K be a field with absolute Galois group $G_K = \mathrm{Gal}(\overline{K}/K)$. Let M be a topological G-module over R.
   The *Galois cohomology group of K with coefficients in M* is:
   $$H^n(K, M) = H^n(\overline{K}/K, M) = H^n_{\mathrm{cont}}(\mathrm{Gal}(\overline{K}/K), M).$$

In either case, one also lets $H^n(G, M)$ denote the Galois cohomology groups where G is the relevant Galois group.

Appendix D. Selmer groups

Definition D.0.1 (Kummer map associated to an isogeny of an abelian variety)**.** Let k be a field, and let A and B be commutative group schemes (Definition A.0.2) over S. Suppose $\varphi : A \to B$ is an isogeny over S. Denote by $G_k = \mathrm{Gal}(\overline{k}/k)$ the absolute Galois group of k.

The short exact sequence

$$0 \to \ker\varphi \to A(\overline{k}) \xrightarrow{\varphi} B(\overline{k}) \to 0.$$

yields the following long exact sequence in Galois cohomology (Definition C.0.5):

$$\begin{aligned} 0 \to \ker\varphi(k) \to & A(k) \xrightarrow{\varphi} B(k) \\ \xrightarrow{\delta_\varphi} H^1(G_k, \ker\varphi) \to & H^1(G_k, A(\overline{k})) \to H^1(G_k, B(\overline{k})) \to \cdots. \end{aligned}$$

The *Kummer map associated to the isogeny φ* is the Galois cohomological connecting homomorphism

$$\delta_\varphi : B(k) \to H^1(G_k, \ker\varphi),$$

above. In fact, there is a short exact sequence

$$0 \to B(k)/\varphi(A(k)) \xrightarrow{\delta'_\varphi} H^1(G_k, \ker \varphi) \to H^1(G_k, A)[\varphi] \to 0$$

where the map δ'_φ is induced by δ_φ. We may also let the *Kummer map associated to φ* refer to this map δ'_φ. We may also use the abuse of notation δ_φ to denote δ'_φ.

Definition D.0.2 (Selmer group for an isogeny of abelian group schemes over a global field)**.** (♠ TODO: It should be possible to define this for more general group schemes) Let K be a global field, and let A and B be abelian group schemes (Definition A.0.2) defined over K. Suppose $\varphi : A \to B$ is an isogeny defined over K.

Denote by $G_K = \mathrm{Gal}(\overline{K}/K)$ the absolute Galois group of K. For each place v of K, fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group (Definition E.0.3) $G_v \subset G_K$.

The *Selmer group of φ over K*, denoted by notations such as $\mathrm{Sel}(A/K)$, $\mathrm{Sel}^\varphi(A/K)$, $\mathrm{Sel}^{(\varphi)}(A/K)$, $\mathrm{Sel}_\varphi(A/K)$, is the subgroup of the Galois cohomology group (Definition C.0.5) $H^1(K, \ker \varphi)$ given by

$$\mathrm{Sel}^\varphi(A/K) := \ker\left(H^1(G_K, A[\varphi]) \to \prod_v H^1(G_v, A(\overline{K}))[\varphi]\right),$$

(♠ TODO: define the kummer map associated to φ) where the product runs over all places v of K, and $H^1(G_v, A)[\varphi]$ denotes the image of the local Kummer map associated to φ.

We describe the map

$$H^1(G_K, A[\varphi]) \to \prod_v H^1(G_v, A(\overline{K}))[\varphi] \tag{A}$$

(♠ TODO: define a decomposition group) used to define the kernel above: Note that G_v acts on $A(\overline{K}_v)$ and $B(\overline{K}_v)$, and the base change φ_v of φ to K_v induces a Kummer short exact sequence (Definition D.0.1)

$$0 \to B(K_v)/\varphi(A(K_v)) \xrightarrow{\delta_{\varphi_v}} H^1(G_v, A[\varphi]) \to H^1(G_v, A(\overline{K}_v))[\varphi] \to 0.$$

The natural inclusions $G_v \subset G_K$ and $E(K) \subset E(K_v)$ give restriction maps $H^1(G_K, A[\varphi]) \to H^1(G_v, A[\varphi])$ on Galois cohomology, and we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & B(K)/\varphi(A(K)) \arrow[r, "\delta_\varphi"] \arrow[d] & H^1(G_K, A[\varphi]) \arrow[r] \arrow[d, "\prod_v \mathrm{res}_v"] \arrow[dr, dotted] & H^1(G_K, A(\overline{K}))[\phi] \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \prod_v B(K_v)/\varphi(A(K_v)) \arrow[r, "\delta_{\varphi_v}"] & \prod_v H^1(G_v, A[\varphi]) \arrow[r] & \prod_v H^1(G_v, A(\overline{K}_v))[\phi] \arrow[r] & 0
\end{tikzcd}$$

The map (A) is the one given in the above commutative diagram by the dotted arrow.

Equivalently,

$$\mathrm{Sel}^\varphi(A/K) = \left\{c \in H^1(K, A[\varphi]) : \text{ for all places } v \text{ of } K, \mathrm{res}_v(c) \in \mathrm{Im}\, \delta_{\varphi_v}\right\}$$

Appendix E. Étale sheaves

Theorem E.0.1 (See [?, Tags 0DV5, 0GIY]). (♠ TODO: Work out a statement for $\mathbb{Q}_\ell$-coefficients) Let X be a connected scheme and let $\overline{x} \in X$ be a geometric point.

1. There is an equivalence of categories (♠ TODO: define finite in this context)
$$\left\{\begin{matrix}\text{finite locally constant} \\ \text{sheaves of sets on } X_{\acute{e}tale}\end{matrix}\right\} \longleftrightarrow \left\{\text{finite } \pi_1^{\text{ét}}(X, \overline{x})\text{-sets}\right\}.$$
2. There is an equivalence of categories
$$\left\{\begin{matrix}\text{finite cardinality locally constant} \\ \text{sheaves of abelian groups on } X_{\acute{e}tale}\end{matrix}\right\} \longleftrightarrow \left\{\text{finite cardinality} \pi_1^{\text{ét}}(X, \overline{x})\text{-modules}\right\}.$$
3. For a finite ring Λ, there is an equivalence of categories
$$\left\{\begin{matrix}\text{finite type, locally constant} \\ \text{sheaves of } \Lambda\text{-modules on } X_{\acute{e}tale}\end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix}\text{finite } \pi_1^{\text{ét}}(X, \overline{x})\text{-modules endowed} \\ \text{with commuting } \Lambda\text{-module structure}\end{matrix}\right\}.$$
()
(♠ TODO: in the below, the geometrically unibranch hypothesis might not be needed, but it was in the stacks project)
4. Assume that X is irreducible and geometrically unibranch. For a ring Λ, there is an equivalence of categories
$$\left\{\begin{matrix}\text{finite type, locally constant} \\ \text{sheaves of } \Lambda\text{-modules on } X_{\acute{e}tale}\end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix}\text{finite type} \Lambda\text{-modules } M \text{ endowed} \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \overline{x})\text{-action}\end{matrix}\right\}.$$

Theorem E.0.2. Let (R, λ) be a complete DVR such that R/λ^n is finite for all $n \geq 1$. Let E be the fraction field of R. Let X be a connected noetherian scheme. Let $\overline{x} \in X$ be a geometric point.

1. There is an equivalence of categories between:
$$\left\{\text{lisse } \lambda\text{-adic sheaves on } X_{\acute{e}tale}\right\} \longleftrightarrow \left\{\begin{matrix}\text{finitely generated free } R\text{-modules } M \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \overline{x})\text{-action}\end{matrix}\right\}.$$
2. There is an equivalence of categories between:
$$\left\{\text{lisse objects of } \mathrm{Sh}(X, E)\right\} \longleftrightarrow \left\{\begin{matrix}\text{finite-dimensional } E\text{-vector spaces } V \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \overline{x})\text{-action}\end{matrix}\right\}.$$
3. Let K be an algebraic extension of E. There is an equivalence of categories between:
$$\left\{\text{lisse objects of } \mathrm{Sh}(X, K)\right\} \longleftrightarrow \left\{\begin{matrix}\text{finite-dimensional } K\text{-vector spaces } V \\ \text{with a continuous } \pi_1^{\text{ét}}(X, \overline{x})\text{-action}\end{matrix}\right\}.$$

Definition E.0.3. Assume **??** and further assume that L/K is a Galois extension.

The *decomposition group of L/K* is the subgroup
$$D_w = \{\sigma \in \mathrm{Gal}(L/K) : w \circ \sigma = w\}.$$

It is also the inverse limit of the decomposition groups of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of $\mathrm{Gal}(L/K)$.

Letting (K, v) be a valuation field, its *decomposition group* $D_{K,v}$ (also sometimes denoted by $G_{K,v}$) is the decomposition group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition E.0.4. Assume **??** and further assume that L/K is a Galois extension.

There is a natural surjective homomorphism

$$D_w \to \mathrm{Gal}(k_w/k_v)$$

(Definition E.0.3) on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_v} \text{ for all } x \in \mathcal{O}_v\}.$$

(Definition E.0.3) It is also the inverse limit of the decomposition groups of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of D_w.

Letting (K, v) be a valuation field, its *inertia group* $I_{K,v}$ is the inertia group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Proposition E.0.5 (see [?, Tag 0BQM])**.** Let X be a normal integral scheme function field K. Let η be the generic point of X and let $\overline{\eta}$ be a geometric point of X whose image is η. The canonical continuous group homomorphism

$$\mathrm{Gal}(K^{\text{sep}}/K) \cong \pi_1^{\text{ét}}(\eta, \overline{\eta}) \to \pi_1(X, \overline{\eta})$$

(♠ TODO: better describe the canonical map)

is identified with the quotient map

$$\mathrm{Gal}(K^{\text{sep}}/K) \to \mathrm{Gal}(M/K)$$

(♠ TODO: defined unramifiedness of X in L) where $M \subseteq K^{\text{sep}}$ is the union of the finite subextensions L such that X is unramified in L.